

MAT2041 Assignment 7: Singular Value Decomposition and Quadratic Forms

Rules (please read!!)

- **English:** Answer the questions in English. Otherwise, you will lose half of the points.
- **Electronic submission:** Turn in solutions electronically via Blackboard. Be sure to submit your homework as a single file.
- **Collaboration policy:** Collaboration is allowed for all problems, but please list all the people with whom you discussed. Crediting help from other classmates will not take away any credit from you.

However, only insightful discussions are allowed. Directly sharing the solutions is prohibited. The details of the collaboration policy for this course are available in the Resources tab on Piazza.

- **Questions on homework:** Start early and come to TA office hours with your questions on the assignments!

Total: 150 points, 7 problems.

1. Judgement (30 points)

State true or false. If true, provide brief reasons. If false, provide a counter-example.

- (a) Any real square matrix A can be written as $A = SDS^T$, where D is a diagonal matrix, and S is an orthogonal matrix.
- (b) If A and B are similar, then $\det(A) = \det(B)$.
- (c) Any n by n real matrix A has at least one real eigenvalue.
- (d) Any nonzero real matrix A has at least one nonzero singular value.
- (e) For any real matrix A , $A^T A$ and AA^T have the same multiset of eigenvalues.
- (f) If A and B have the same multiset of eigenvalues, then A and B are similar.
- (g) Any rank r matrix can be written as the sum of r rank-one matrices.
- (h) In the full SVD $A = U\Sigma V^T$, the column space of A is the same as the column space of U .
- (i) If all entries of a real symmetric matrix are nonnegative, then it is positive semidefinite.
- (j) Suppose A is a real symmetric $n \times n$ matrix. Then A is positive definite if and only if it can be written as $A = B^T B$, where B is a full-rank $n \times n$ matrix.

2. Full SVD (20 points)

Find a full SVD (singular value decomposition) of each of the following matrices:

Remark: You need to show all intermediate steps.

(a) $\begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ -1 & 1 \end{bmatrix}$

3. Minimum Norm Solution (20 points)

Let A be a matrix with full row rank r .

- (a) Show that the full SVD of A can be written as

$$A = U \begin{bmatrix} D & 0 \end{bmatrix} V^T,$$

in which D is an invertible diagonal matrix of order r .

- (b) Show that if $Ax = b$, then

$$\|x\| \geq \|D^{-1}U^T b\|.$$

Hint: Write x as $x = \sum_{j=1}^n a_j v_j$ in $Ax = b$, and consider the first r entries of this sum. Here v_j is the j -th column of V .

4. Pseudo-Inverse (20 points)

Let A be an $m \times n$ matrix with $m \geq n$ and $\text{rank}(A) = n$. Start with the full singular value decomposition

$$A = U\Sigma V^T,$$

where

$$\Sigma = \begin{bmatrix} D \\ 0 \end{bmatrix} \in \mathbb{R}^{m \times n},$$

and

$$D = \text{diag}(\sigma_1, \dots, \sigma_n)$$

with $\sigma_1, \dots, \sigma_n > 0$.

Let Σ^+ denote the $n \times m$ matrix

$$\Sigma^+ = \begin{bmatrix} \frac{1}{\sigma_1} & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & \frac{1}{\sigma_2} & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \frac{1}{\sigma_n} & 0 & \cdots & 0 \end{bmatrix}.$$

Define

$$A^+ = V\Sigma^+U^T.$$

(a) Show that

$$A^+A = I_n.$$

Note that A^+ is called the pseudo-inverse of A .

(b) Show that $\hat{x} = A^+b$ satisfies the normal equation

$$A^T Ax = A^T b.$$

5. Trace Identity (10 points)

Let A be an $m \times n$ matrix and B be an $n \times m$ matrix. Prove that

$$\operatorname{tr}(AB) = \operatorname{tr}(BA).$$

Hint: Write out the diagonal entries of AB and BA in terms of the entries of A and B .

6. Rank-1 Decomposition (30 points)

Suppose $x_j \in \mathbb{R}^{m \times 1}$, $j = 1, \dots, k$, are linearly independent, and $y_j \in \mathbb{R}^{n \times 1}$, $j = 1, \dots, k$, are linearly independent. Suppose

$$A = \sum_{j=1}^k x_j y_j^T.$$

In this problem, we will guide you to prove

$$\mathcal{C}(A) = \text{span}(\{x_1, \dots, x_k\}).$$

- (a) Prove the case of $k = 1$, i.e., if $A = uv^T$ for nonzero vectors u, v , then

$$\mathcal{C}(A) = \text{span}(\{u\}).$$

- (b) Prove

$$\mathcal{C}(A) \subseteq \text{span}(\{x_1, \dots, x_k\}).$$

Hint: There are multiple ways to prove this. You can use the matrix form, or write columns of A in terms of the x_j 's and entries of the y_j 's.

- (c) Prove

$$\text{span}(\{x_1, \dots, x_k\}) \subseteq \mathcal{C}(A).$$

Hint: One way to prove this is to use a matrix form and solvability of a linear system with full row/column rank coefficient matrices in Lec 14.

7. Shape of a Conic Section (20 points)

Determine the shape of the conic section

$$4x_1^2 + 4x_1x_2 + 7x_2^2 = 6.$$

Transform it to standard form by using an orthogonal change of variables.